

Mohamed Ahmed

ahme0712@umn.edu | (763) 248-8763 | [linkedin.com/in/mohamed-ahmed-89b807328](https://www.linkedin.com/in/mohamed-ahmed-89b807328)

Education

University of Minnesota

Expected May 2027

Bachelor of Science in Electrical Engineering

Relevant Coursework: Power Electronics, Control Systems, Analog Electronics, Embedded Systems, Wireless Hardware Design, Antenna Design

Experience

Michels Power

May 2026 – Aug 2026

Project Management Engineering Intern

- Supported electric utility projects through field verification, GIS records, engineering drawings, and technical documentation.
- Reviewed engineering drawings and as-built records for transmission and distribution projects.
- Developed a technical database tracking **1,200+ utility as-built records** for project documentation.

Club Experience

STRIVE Medical

September 2025 – Present

Electrical / Hardware Team Lead

- Lead electrical development for a **medical robotics platform** integrating motor-drive hardware, embedded control, sensing, and power distribution.
- Designed a **4-layer PCBA** supporting power distribution, embedded controllers, sensor interfaces, and motor-drive hardware.
- Integrated **Teensy 4.1, ESP32, motor controllers, IMUs, and 14-bit encoders** across robotic subsystems.
- Developed and debugged motor-control and sensor interfaces using oscilloscopes, logic analyzers, and embedded firmware.
- Led electrical development to **unanimous 1st Place** at the 2026 UMN MVP Challenge.

Projects

100 W Synchronous Buck Converter | Power Electronics, PCB Design, Digital Control

- Designed a **100 W, 24 V-to-12 V synchronous buck converter** using N-channel MOSFETs, gate-drive circuitry, voltage/current sensing, and protection hardware.
- Modeled converter operation in **MATLAB** using averaged and small-signal models to derive plant dynamics and evaluate closed-loop behavior.
- Performed extensive **LTspice simulation** of the switching power stage, sensing circuitry, transient response, and component behavior prior to hardware implementation.
- Designed a **4-layer power PCB** with attention to high-current switching loops, gate-drive paths, grounding, sensing, and separation of power and control circuitry.
- Designed a **PI voltage-control loop** using Bode plots and stability margins, with STM32G4 PWM generation and ADC-based feedback for digital control.

Wearable PPG Monitoring System | Power Management, Mixed-Signal Hardware, Embedded Systems

- Designed a mixed-signal wearable PCBA integrating regulated power, an optical analog front end, **24-bit ADC, IMU, ESP32-C3**, and BLE.
- Developed power and signal interfaces supporting low-noise optical sensing and battery-powered embedded operation.
- Designed analog sensing circuitry and integrated analog, digital, and power subsystems on a custom PCB.
- Performed circuit simulation, PCB design, and hardware/software debugging across the sensing and embedded system.

Skills

Power Electronics: DC-DC Converters, Synchronous Buck Converters, MOSFETs, Gate Drivers, PWM, Voltage/Current Sensing

Modeling & Controls: Small-Signal Modeling, Transfer Functions, Bode Analysis, Stability Margins, PI Control, MATLAB

PCB & Hardware: Power PCB Design, 4-Layer PCBs, Switching-Loop Layout, Grounding, Sensor Interfaces, PCB Bring-Up

Embedded Control: STM32G4, ESP32, Teensy 4.1, Embedded C/C++, ADC, PWM, SPI, I2C, UART

Debugging & Test: Oscilloscope, Logic Analyzer, Multimeter, LTspice, Hardware Debugging

Tools: MATLAB, LTspice, Altium Designer, KiCad, Git, SolidWorks